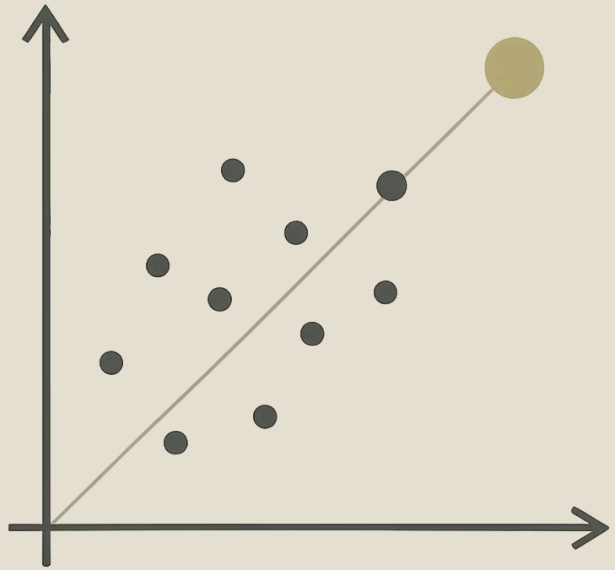


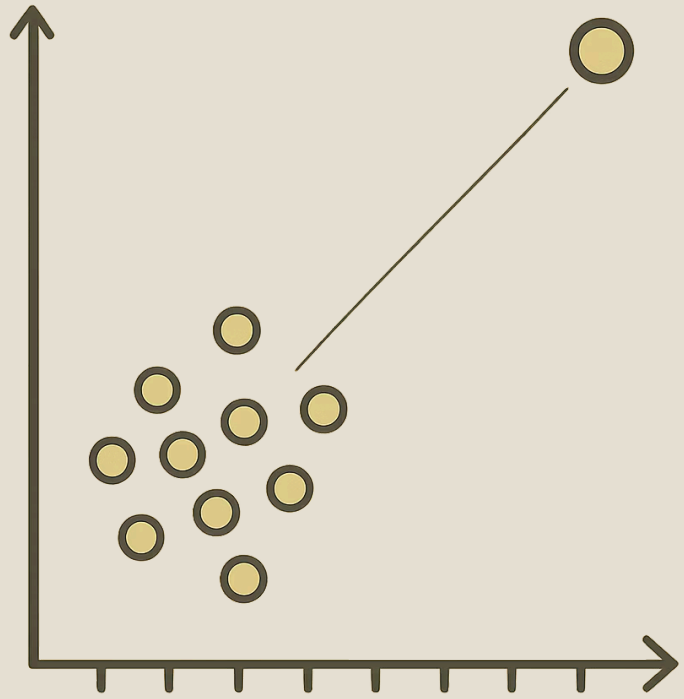
Outlier Detection and Data Mining Applications

Unit Topic: Outliers and Data Mining Applications

PREPARED BY DIVYA TRIPATHI

RA2311003030547

BTECH CSE CORE - I



What Is an Outlier?

An **outlier** is a data point that differs significantly from the rest of the dataset. It may signal errors, rare events, or unusual behavior — and identifying them is essential for maintaining data quality and model accuracy.

Measurement Errors

Faulty sensors or instruments produce incorrect readings.

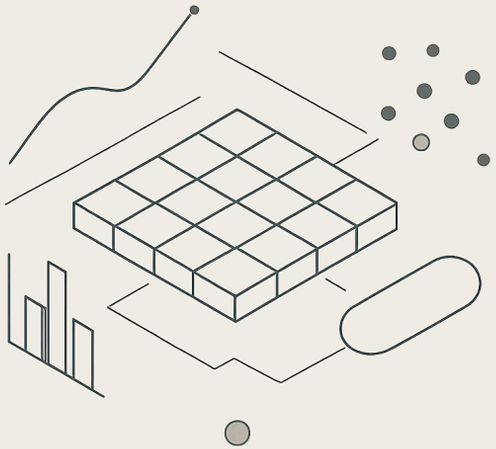
Data Entry Mistakes

Human errors during manual input create anomalies.

Genuine Rare Events

Real but unusual occurrences worth investigating.

Challenges in Detecting Outliers



Outlier detection becomes increasingly difficult as data grows in size, complexity, and velocity. Four key challenges stand out:

Large Dataset Size

Processing massive datasets demands significant computational resources.

High Dimensionality

Outliers become harder to identify as the number of features grows.

Noise in Data

Random noise can be mistaken for true outliers.

Dynamic Data Streams

Real-time systems require continuous, adaptive detection.

Outlier Detection Methods

Outlier detection approaches are classified by how much labeled data they require. The choice depends on data availability and the problem context.

Supervised

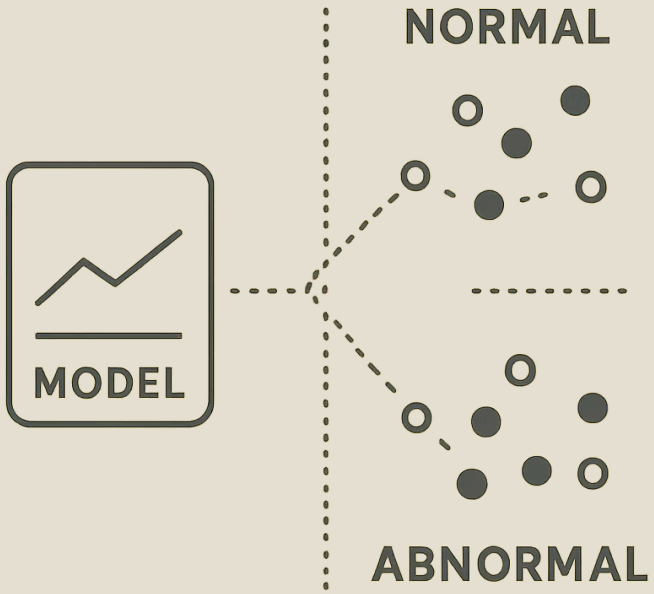
Uses fully labeled normal and abnormal data.

Semi-Supervised

Uses mostly normal labeled data; detects deviations.

Unsupervised

No labels required; detects outliers via patterns and distances.



Supervised Outlier Detection

Supervised methods train classification models on **fully labeled datasets** containing both normal and abnormal examples. When sufficient labeled data exists, these methods achieve high accuracy.

Fraud Detection

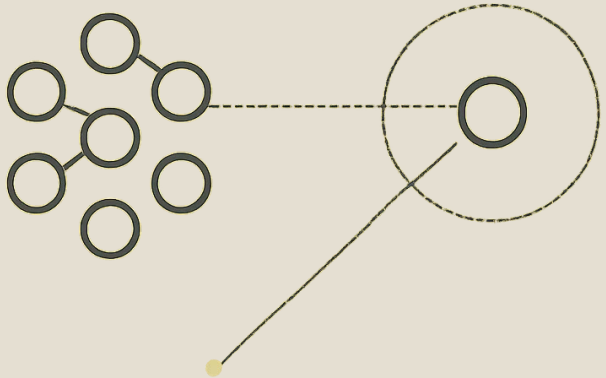
Identifies fraudulent transactions in banking systems.

Spam Detection

Classifies unwanted emails using labeled training data.

Medical Diagnosis

Detects abnormal patient data indicative of disease.



CLUSTERING

UNSUPERVISED OUTLIER DETECTION

Semi-Supervised & Unsupervised Detection

Semi-Supervised Methods

Trains on **labeled normal data only**, flagging deviations as outliers. Ideal when outlier examples are rare or unavailable.

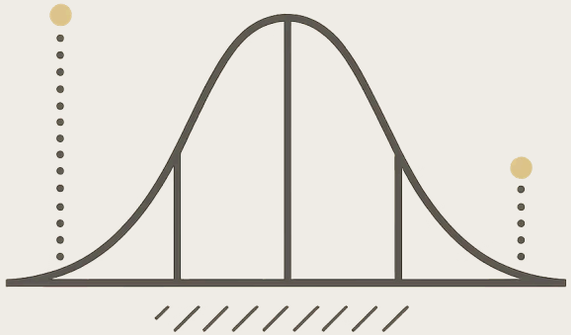
- Requires only normal-class labels
- Detects deviation from learned norms
- Useful when anomalies are scarce

Unsupervised Methods

Requires **no labeled data**. Outliers are identified through intrinsic data structure.

- **Distance-based:** Points far from neighbors
- **Density-based:** Points in sparse regions
- **Clustering:** Points outside any cluster

Statistical Data Mining Approaches



Statistical methods are among the most interpretable and widely used techniques for outlier detection, especially with numerical data.

Mean & Standard Deviation

Flags points beyond a threshold number of standard deviations.

Z-Score Method

Measures how many standard deviations a point is from the mean.

Probability Distribution Models

Models data distribution; low-probability points are outliers.

Hypothesis Testing

Statistically tests whether a point belongs to the population.

Data Mining in Recommender Systems

A **recommender system** uses data mining to suggest relevant items to users, improving engagement and personalization.

Behavior Analysis

Analyzes purchase history and browsing patterns to understand user preferences.

Similarity Matching

Identifies similar users or products to generate relevant recommendations.

Predictive Personalization

Predicts user preferences to surface items they are likely to enjoy.

Applications: Online shopping, movie streaming, music platforms, and news feeds.

Data Mining Applications

Intrusion Detection

Data mining monitors network traffic to detect unauthorized access and suspicious activity in real time.

- Detects unauthorized system access
- Identifies anomalous network patterns
- Core to modern cybersecurity systems

Financial Analysis

In finance, data mining uncovers fraud, analyzes spending behavior, and supports risk management decisions.

- Detects fraudulent transactions
- Analyzes customer spending patterns
- Supports risk and investment planning

Thank You

We hope this overview of outlier detection and data mining applications was insightful. Questions and discussions are welcome.